

Vivek Notes For Mounting an EBS Volume to an EC2 Instance

How to format, mount, and use an EBS volume on an EC2 Linux instance

1. Create EBS Volume



2. Attach EBS Volume to EC2



3. Check volume

```
lsblk
```



4. Inspect disks

```
sudo fdisk -l
```



5. Check filesystem

```
sudo file -s /dev/nvme1n1
```



6. If new/raw volume

```
sudo mkfs -t xfs /dev/nvme1n1
```



7. Verify filesystem

```
sudo file -s /dev/nvme1n1
```



8. Create mount point

```
sudo mkdir /mydata
```



9. Mount volume

```
sudo mount /dev/nvme1n1 /mydata
```



10. Use /mydata

Steps For Attach and Mount EBS Volume to EC2

Step 1 – Create EC2 instance

Step 2 — Create an EBS Volume

Go to AWS Console:

EC2 → Elastic Block Store → Volumes → Create volume

Setting	Value
Volume type	gp3
Size	10 GiB
Availability Zone	Same AZ as your EC2
Tags	Name = EBS-Practical

Important

The EBS volume and EC2 instance must be in the same Availability Zone for normal attachment.

For example:

EC2 → ap-south-1a

EBS → ap-south-1a

Step 3 — Attach EBS to EC2

Go to:

EC2 → Volumes

Select your new volume.

Click:

Actions → Attach volume

Select your EC2 instance.

AWS may show a device name such as:

`/dev/sdf`

Click:

Attach volume

Your EBS is now attached.

Created volume

↓ attach with

EC2 Instance

Check the attached disk

Run:

`lsblk`

Important

Do not assume the device name is always `/dev/nvme1n1`.

Look at your actual `lsblk` output.

Your new volume will generally be the disk with the new size you just created

Then Follow This Steps

Here are the commands you'll need to mount your EBS volume to an EC2 instance:

1. Run the command to show the volumes: `lsblk`
2. List the disk partition on the system: `sudo fdisk -l`
3. Check the file system: `sudo file -s /dev/nvme1n1`
4. If the output is data, then create the file system: `sudo mkfs -t xfs /dev/nvme1n1`
5. Recheck the file system: `sudo file -s /dev/nvme1n1`
6. Create a mount point directory for the volume: `sudo mkdir /mydata`

7. Mount the volume: `sudo mount /dev/nvme1n1 /mydata`

After That umount /mydata

And detach that volume from current instance and make another instance and attach that volume to new instance

And follow that commands

Step 1 :- Go To root user

`Sudo su -`

Step 2 :- make directory

`Sudo mkdir /vivek`

Step 3 :- mount that directory with volume

`Sudo mount /dev/nvme1n1 /vivek`

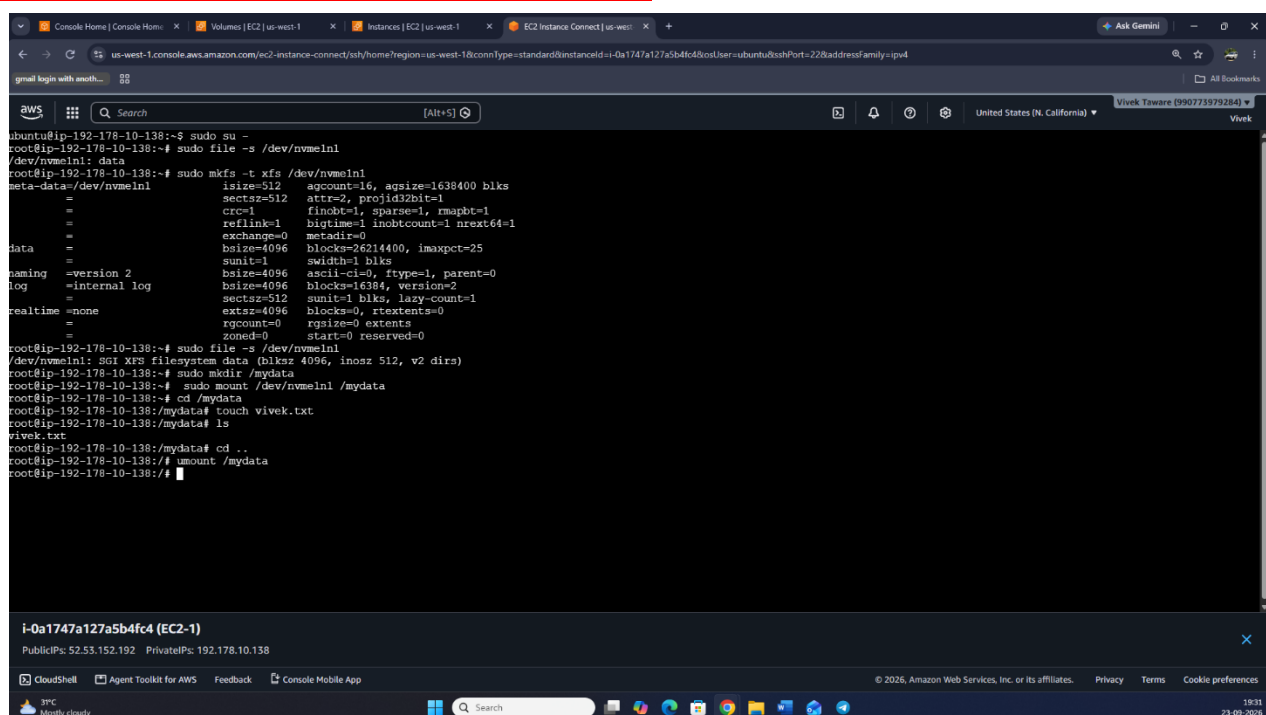
Step 4 :- go to that directory

`cd /vivek`

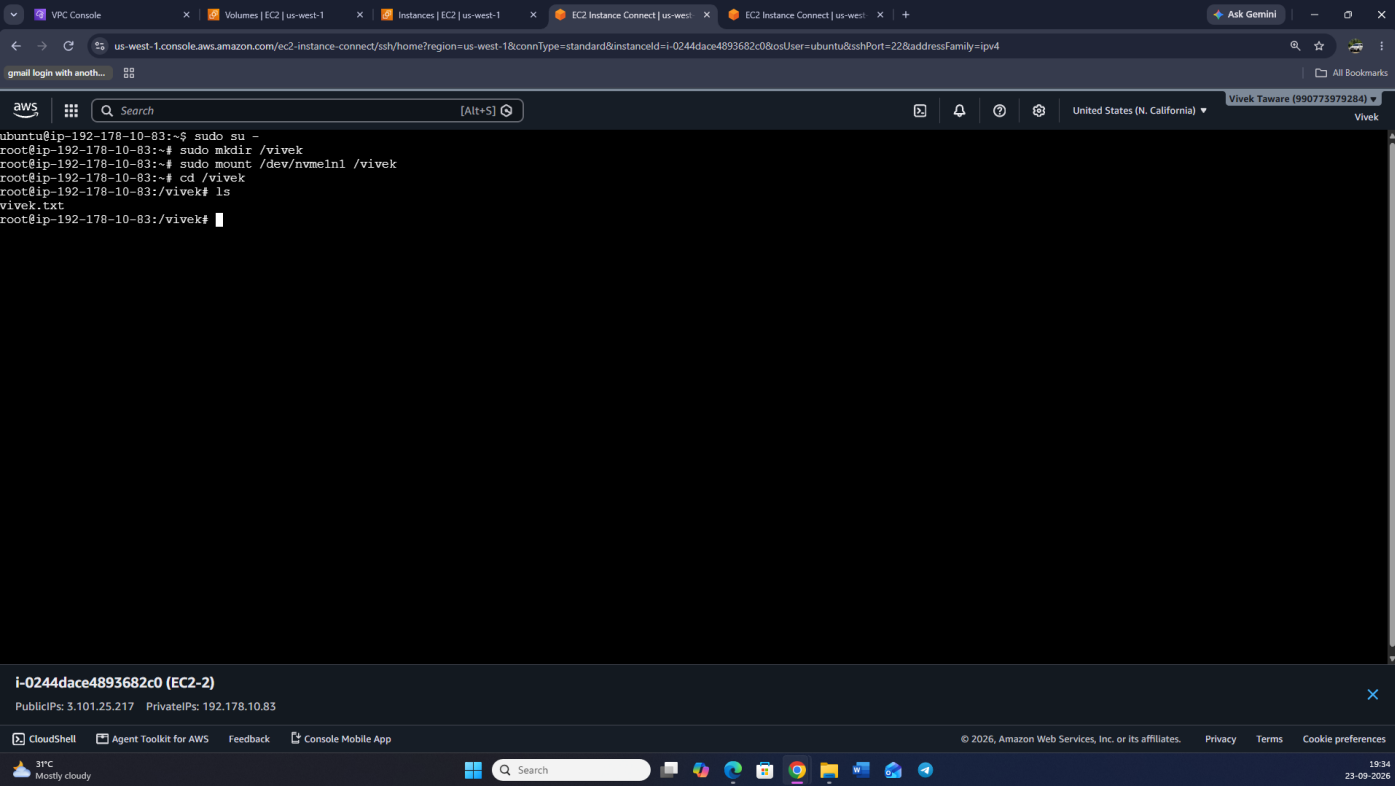
step 5 :- check that data you stored in that volume is available or not

`ls`

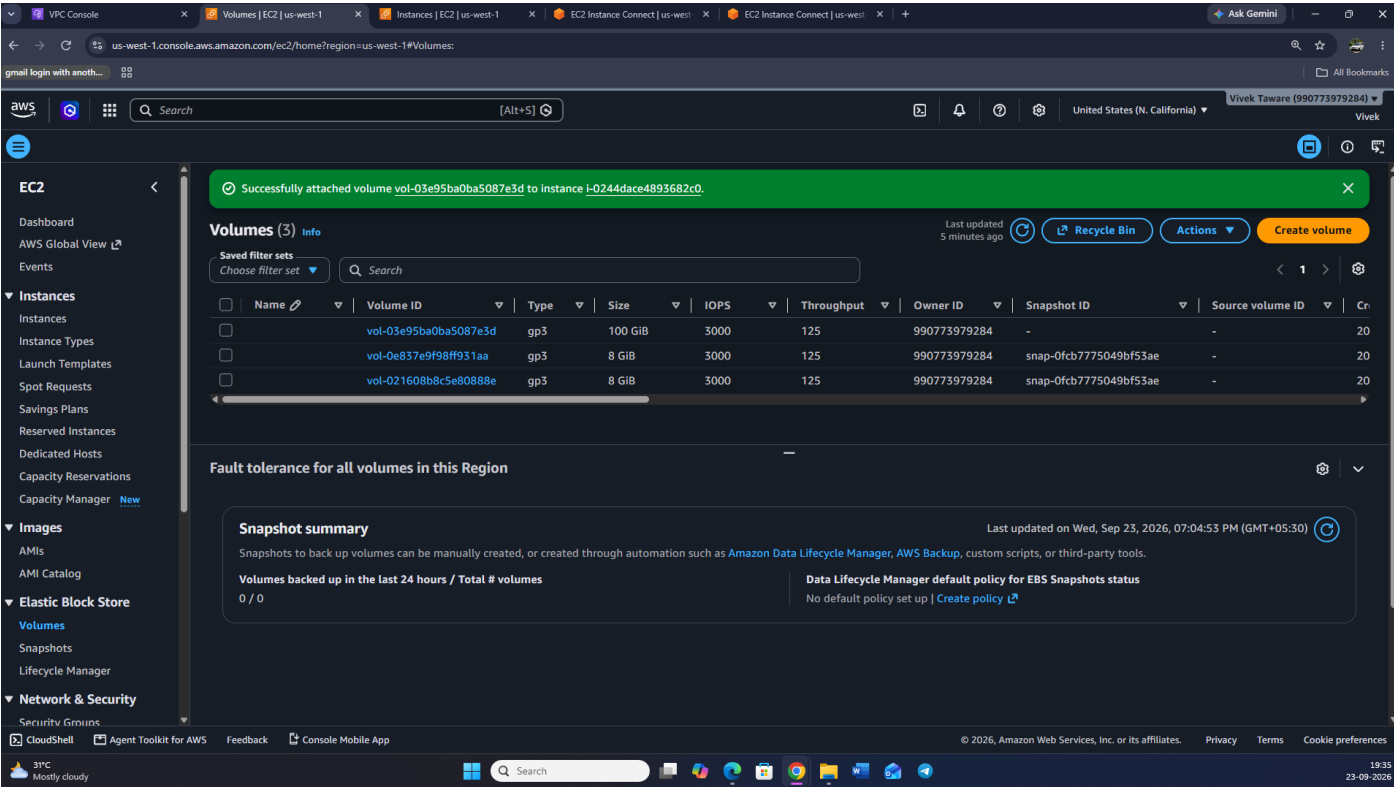
First Instance Mounting Screen Shot



Second instance mounting ScreenShot



Volumes ScreenShot



Term	Meaning	Purpose
EBS	Elastic Block Store	Persistent block storage for EC2
Volume	Virtual disk	Provides storage
Block Device	Disk exposed to Linux	Allows OS to interact with storage
<code>/dev/nvme1n1</code>	Linux device name	Represents your EBS disk
Filesystem	Structure used to store files	Makes disk usable for files
XFS	Linux filesystem	Used to organize/store files
<code>mkfs</code>	Make filesystem	Creates filesystem
Mount	Attach filesystem to directory	Makes storage accessible
Mount Point	Directory where disk is mounted	Example <code>/mydata</code>
<code>lsblk</code>	List block devices	Check attached disks
<code>fdisk -l</code>	List disk/partition information	Inspect disks
<code>file -s</code>	Identify filesystem/data	Check disk contents/type

- What is an EBS Volume?

EBS = Elastic Block Store

An EBS volume is basically a virtual hard disk that you can attach to an EC2 instance

For example:

EC2 Instance

|
| attached

↓

EBS Volume

|
↓

`/dev/nvme1n1`

|
↓

XFS Filesystem

|



/mydata

You can use /mydata to store:

Application files

Database data

Logs

Backups

COMMANDS :-

lsblk

lsblk

Meaning

lsblk = List Block Devices

It shows the disks and partitions available to your Linux EC2 instance.

Example:

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
nvme0n1		259:0	0 8G	0	disk	
--nvme0n1p1		259:1	0 8G	0	part	/
nvme1n1		259:2	0 10G	0	disk	

Here:

nvme0n1 → Root EBS volume

nvme1n1 → Newly attached EBS volume

Use case

Use it when you want to check:

"Is my newly attached EBS volume visible to EC2?"

```
sudo fdisk -l
```

```
sudo fdisk -l
```

Meaning

fdisk is a disk/partition management utility.

-l means list.

It provides more detailed information about disks and partitions.

Example:

Disk /dev/nvme1n1: 10 GiB

Use case

You use it to inspect:

- Disk size
- Partitions
- Disk layout
- Device names

Think:

```
lsblk
```



Quick view

```
fdisk -l
```



Detailed disk/partition information

4.

```
sudo file -s /dev/nvme1n1
```

```
sudo file -s /dev/nvme1n1
```

This is an important command.

Meaning

It checks what type of data/filesystem exists on the disk.

For example:

`/dev/nvme1n1`: data

This generally means the volume doesn't currently contain a recognizable filesystem.

After formatting, you might see something similar to:

`/dev/nvme1n1`: SGI XFS filesystem data

Use case

You use this before formatting to determine whether the volume already has a filesystem.

⚠ Important: Don't blindly run `mkfs` on a volume containing data. Formatting can destroy existing data.

5. `sudo mkfs -t xfs /dev/nvme1n1`

`sudo mkfs -t xfs /dev/nvme1n1`

Meaning

`mkfs` = Make Filesystem

`-t xfs` = create an XFS filesystem

So you're essentially saying:

"Prepare this EBS disk so Linux can store files on it using XFS."

Before:

EBS Volume



Raw disk



No filesystem

After:

EBS Volume



XFS filesystem



Ready to mount

Use case

A newly created EBS volume generally needs a filesystem before you can conveniently use it for file storage.

6. Check again

```
sudo file -s /dev/nvme1n1
```

Now you check whether the filesystem was successfully created.

You should see something indicating:

XFS filesystem

So your process is:

Check



data



Format



XFS

7. Create Mount Point

```
sudo mkdir /mydata
```

Meaning

mkdir = Make Directory

You're creating:

/mydata

This directory becomes the mount point.

Think of it as the location where Linux will make the EBS volume accessible.

For example:

`/mydata`

```
|
|— file1.txt
|— application/
|— logs/
└— backup/
```

Use case

You can choose another directory name too:

```
sudo mkdir /data
```

or

```
sudo mkdir /storage
```

or

```
sudo mkdir /appdata
```

8. Mount the EBS Volume

```
sudo mount /dev/nvme1n1 /mydata
```

This is the step that actually connects the filesystem on the disk to the directory.

Before:

```
/dev/nvme1n1
```

```
/mydata
```

They are separate.

After:

```
/dev/nvme1n1
```



mount



/mydata

Now:

```
cd /mydata
```

and files you create there are stored on the EBS volume.

For example:

```
cd /mydata
```

```
sudo touch test.txt
```

```
ls
```

You should see:

```
test.txt
```

Note : All commands are highlighted with green colour

Description or information is Highlighted with pink colour